Name: _____ Date: _____ Class: _____

MyDesign Portfolio

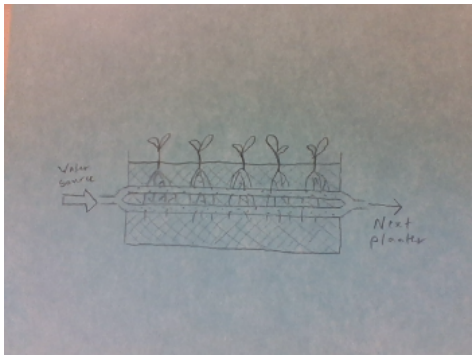
Element D Initial Template

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

Design Idea #1 - ***Under Plant Irrigation***

Date: 3/5/2025
Description: **The water filtration system will be pipes coming from under the plants and straight into the soil, this way it can be spread around properly. The material used will consist of non molding properties that way the pipe doesn't mold and cause bacteria to spread in the air, and on the plants.**
Drawing that includes distinguishing information such as scale, dimensions, and materials:

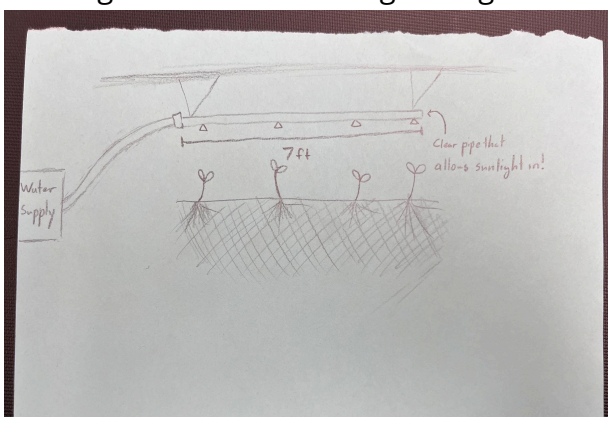


Design Idea #2 - ***Sunshine Pipe***

Date: 3/5/2025
Description: **The watering system will consist of a clear material or tubing that can allow sunlight to shine within the pipe. This allows for less shaded areas and allows people to see inside the watering system for any mold. Mold is known to grow in shaded areas and so a clear system that has sunlight shining in will not allow for mold to grow which**

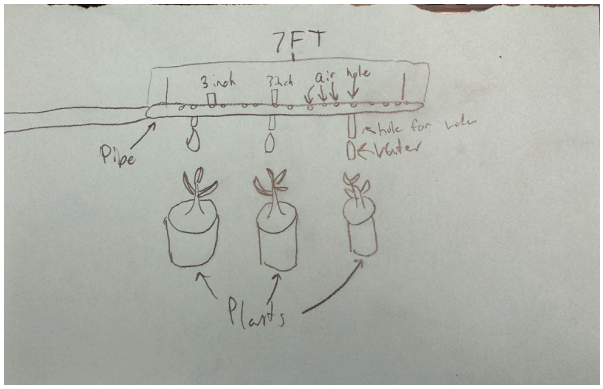
Name: _____ **Date:** _____ **Class:** _____

Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea #3 ***- Air Hole Pipe***

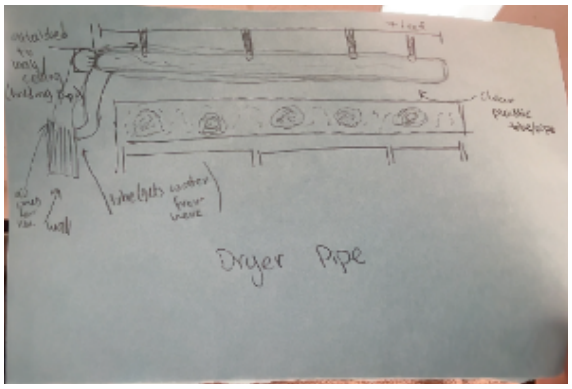
Date: 3/5/2025
Description: **The watering system will consist of small air on the top, and water flowing through the bottom, that way the pipe can dry within each water timer, minimizing the amount of mold that would be made.**
Drawing that includes distinguishing information such as scale, dimensions, and materials:



Name: _____ **Date:** _____ **Class:** _____

Design Idea #4 ***Dryer Pipe***

Date: 3/5/2025
Description: **The watering system will consist of a machine that airs and dries the pipe to remove moisture and chances of mold.**
Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea Comparison and Selection *(D1, D2)*

	weight	***Under Plant Pipe***	***Sunshine Pipe***	***Air Hole Pipe***	***Dryer Pipe***
Non-Moldy	3	2	3	3	3
Sturdy	2	2	2	1	2
How it looks	1	1	3	2	2
Easy to build	2	1	3	2	1
Distributes water	2	2	2	1	1
score	10	8	13	9	9

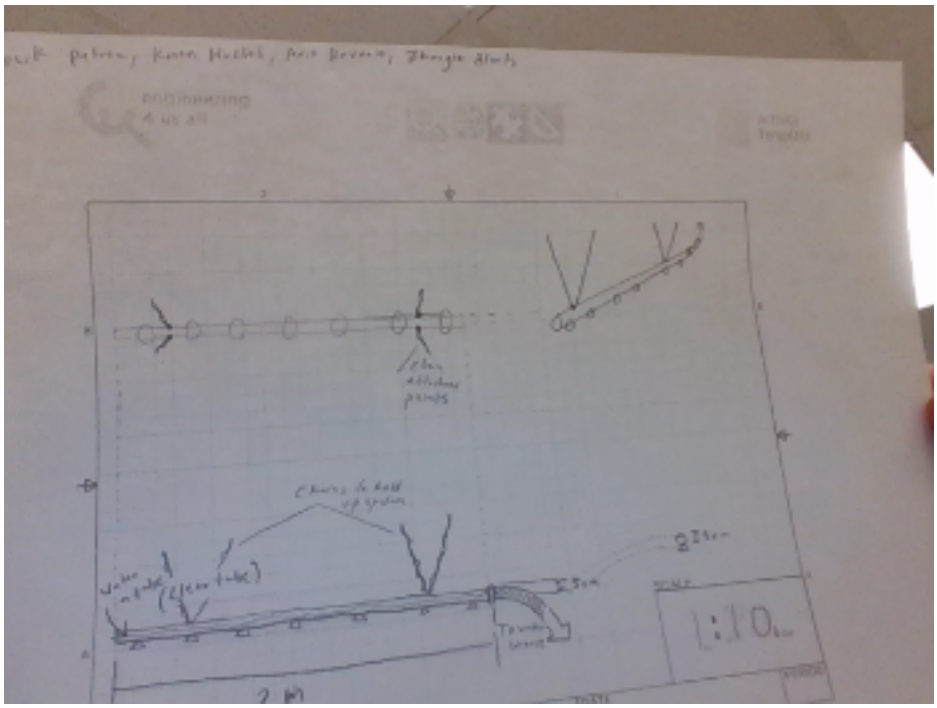
Name: _____ **Date:** _____ **Class:** _____

Our team chose to design <u>Sunshine Pipe</u> to solve the community problem because it is easy to build, it is made of a sturdy material, it is the most effective in not molding and it looks clean and neat.

Design Blueprint *(D4)*

Design Blueprint #1

Date: 3/7/2025
Sketch or CAD Drawing:



Manufacturing Methods:

Step 1: Connect the water pipe to the water system, and make sure the water gets through the pipe

Step 2 : Hang the pipe to the ceiling, making sure of the correct height.

Step 3 : Turn on, make sure the water is going on the plants and evenly. Finally checking after 2 weeks and then a month to see if the tube is molding.

Name: _____ **Date:** _____ **Class:** _____

Supply List and Costs:

Item	**Cost each**	**# needed**	**Total**
Instant mold and Mildew resistant spray	29.99	1 needed	29.99
7 feet silicon pipe	22.10	1 seven foot long	22.10
Water connector	9.36	1 or 2 needed	18.72
Total Cost	61.45		70.81

Plan of Action *(D3)*

Link :

Iterations for the Solution: